

Mathematical Formulation of the Model

Sets

- period: Set of periods p
- month: Set of months m
- time: Set of time steps t

Parameters

- $E_{\text{cons}}[t]$: Energy consumption at time t
- $E_{\text{prod}}[t]$: Produced energy at time t
- $TP[p]$: Power tariff for period p for the entire year
- $TE[m][p]$: Energy tariff for month m and period p
- $Time[p]$: Time steps in period p
- tep : Penalization tariff coefficient
- $Kp[p]$: Penalty coefficient for period p
- Δt : Time step duration
- TB : Battery tariff depending on the capacity
- $TB_{\text{maintenance}}$: Tariff for the maintenance of the batterie per year
- N_{days} : Number of day for the optimization
- $Effc$: Efficiency for charging the batterie
- $Effd$: Efficiency for discharging the batterie

Variables

- $P_{\text{prev}}[p]$: Power contracted for the year p
- $P_c[t]$: Charging power at time t
- $P_d[t]$: Discharging power at time t
- $E[t]$: Energy stored in the battery at time t
- C_b : Capacity of the batterie
- $E_{\text{grid}+}[t]$: Energy consumed from the grid at time t
- $E_{\text{grid}-}[t]$: Energy injected into the grid at time t

- $P_+[t, p]$: Positive power difference between the consumed power and the contracted one at time t and period p
- $P_-[t, p]$: Negative power difference between the consumed power and the contracted one at time t and period p

Objective Function

Minimize the total cost:

The total cost is composed of the cost for the contracted power, the cost of the electricity energy from the grid, the cost of the penalisation and the battery cost.

$$\begin{aligned}
\text{Minimize } & \sum_p TP[p] \cdot P_{\text{prev}}[p] \cdot N_{\text{days}} \\
& + \sum_{m,p,t} TE[m][p] \cdot E_{\text{grid}+}[t] \\
& + \sum_p Kp[p] \cdot \text{tep} \cdot \sqrt{\sum_t (P_{\text{plus}}[t, p])^2} \\
& + \left(\frac{C_b \cdot TB}{\text{batterie lifetime}} + C_b \cdot TB_{\text{maintenance}} \right) \cdot \frac{N_{\text{days}}}{365}
\end{aligned}$$

Constraints

1. Battery Capacity Constraints:

$$\begin{aligned}
E[t] & \leq C_b \\
E[t] & \geq 0.2 \cdot C_b
\end{aligned}$$

2. Battery Power Constraints:

$$\begin{aligned}
P_c[t] & \leq \text{charge_rate} \cdot C_b \\
P_d[t] & \leq \text{decharge_rate} \cdot C_b
\end{aligned}$$

3. Battery Energy Balance:

$$E[t] = E[t-1] + (\text{Effc} \cdot P_c[t] - \frac{P_d[t]}{\text{Effd}}) \cdot \Delta t$$

4. Grid Energy Balance:

$$E_{\text{grid}+}[t] - E_{\text{grid}-}[t] = E_{\text{cons}}[t] + (P_c[t] - P_d[t]) \cdot \Delta t - E_{\text{prod}}[t]$$

5. Power difference for penalization:

$$P_{\text{plus}}[t, p] - P_{\text{minus}}[t, p] = \frac{E_{\text{cons}}[t] - E_{\text{prod}}[t]}{\Delta t} + (P_c[t] - P_d[t]) - P_{\text{prev}}[p]$$

6. Power contracted constraint:

$$\begin{aligned}P_{\text{prev}}[p] &\leq P_{\text{prev}}[p+1] \\ P_{\text{prev}}[0] &\geq 0\end{aligned}$$